c) $\forall x \forall y\,(xy \geq x)$

We know, $-12 = 4 \times -3 \geq 4$ is a False statement

By defn. of universal quantifier, $\forall y\,(4xy \geq 4)$ is False.

By defn. of universal quantifier, $\forall x \forall y\,(xy \geq x)$ is False.

40) Find a counterexample, if possible, to these universally quantified statements, where the domain for all variables consists of all integers.

a) $\forall x \exists y\,(x = 1/y)$

$2 = 1/(1/2)$. $1/2$ is the only number which when divided by 1 gives 2. $1/2$ is not an integer. $\therefore \exists y\,(2 = 1/y)$ is False, from the defn. of existential quantifier. By the defn. of universal quantifier, $\forall x \exists y\,(x = 1/y)$ is False.

b) $\forall x \exists y\,(y^2 - x < 100)$. Let it be False. By defn. of negation, $\neg \forall x \exists y\,(y^2 - x < 100)$ is True. By ~~defn. of~~ De Morgan's law of quantifiers, $\exists x \neg \exists y\,(y^2 - x < 100)$ is True. By De Morgan's law of quantifiers, $\exists x \forall y \neg (y^2 - x < 100)$ is True. $\therefore \exists x \forall y\,(y^2 - x \geq 100)$.

By defn. of existential quantifier, there exists some element a in the domain such that $\forall y\,(y^2 - a \geq 100)$ is True. By defn. of universal quantifier, for all elements b in the domain, $(b^2 - a \geq 100)$ is True. $\boxed{\therefore b^2 \geq a + 100}$

$\therefore$ If $a = -100$, $b^2 \geq -100 + 100 = 0$

For all integers b, $b^2 \geq 0$ is True. $\therefore$ It holds. $\boxed{\therefore \forall x \exists y\,(y^2 - x < 100) \text{ is False}}$

c) $\forall x \forall y\,(x^2 \neq y^3)$ $\quad y = 4 \quad x = 8 \quad 64 = 8^2 = 4^3$

$\therefore 8^2 = 4^3$ is True. By defn. of universal quantifier, $\forall y\,(8^2 \neq y^3)$ is False. By defn. of universal quantifier, $\forall x \forall y\,(x^2 \neq y^3)$ is False.